

Application of Fuzzy Logic for Collaborative Robot Control

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Abstract

Collaborative robots (cobots) play a crucial role in modern industry by ensuring safe and efficient human–robot interaction. However, traditional control methods struggle with uncertainty handling and adaptability to dynamic environments. This review explores the application of fuzzy logic as a promising approach for cobot control. The article discusses the fundamental principles of fuzzy logic, its advantages over classical methods, and successful case studies. It analyzes current research, including hybrid methods combining fuzzy logic with machine learning and evolutionary algorithms. The paper also highlights existing challenges and potential future research directions. The conclusions emphasize the potential of fuzzy logic to enhance cobot adaptability and reliability in real-world conditions.

Keywords: adaptive control; collaborative robots; fuzzy control; fuzzy logic; human–robot interaction; robot control

1. Introduction

Modern trends in robotics focus on developing mechanisms capable of interacting safely and efficiently with humans in real-time [1,2]. These mechanisms, known as collaborative robots (cobots), are widely used in manufacturing, healthcare, and service [3–5]. Cobots are designed to operate in dynamic environments and must adapt to various uncertainties. They have an advantage over traditional industrial robots because their control systems take into account the variability of human movements, unpredictable external and internal influences, and changing operational parameters [1,6].

Currently, traditional control methods for cobots, such as PID control and model-based approaches, are well-developed [7–9]. However, these methods require precise mathematical models of the system, which poses challenges due to the unpredictability of human behavior. For instance, non-standard operator movements or sudden changes in a cobot's working parameters can destabilize the control system [10–12]. As a result, rigid control algorithms are not always capable of adapting to such changes, which significantly threatens both interaction safety and cobot efficiency [10,11].

Thus, there is a need to develop adaptive control systems that can respond to changing conditions. One of the most promising control methods for cobots is fuzzy logic [13–15]. Unlike classical approaches, fuzzy logic does not require an exact mathematical model of the system and allows for decision-making based on incomplete or uncertain data, mimicking human reasoning. This makes it particularly useful for cobot control, where human behavior is difficult to fit into predefined patterns. By utilizing linguistic variables and fuzzy “if–then” rules, fuzzy logic constructs an adaptive control system that can function effectively even under uncertainty [13,16].



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At present, fuzzy logic is being actively applied in cobot control [16,17]. It is used for adaptive speed regulation, optimizing interaction forces with humans, and enhancing workplace safety [18–20]. For example, fuzzy control algorithms enable cobots to avoid collisions and ensure smooth manipulator motion by adjusting operating parameters in real time [21,22]. Additionally, fuzzy logic helps determine interaction forces with humans, considering individual user characteristics [23].

Despite the problems that fuzzy logic has successfully addressed, several challenges remain that require further analysis [24,25]. One major difficulty is integrating fuzzy logic with other artificial intelligence and machine learning methods to improve adaptability [26–28]. Another challenge is the high computational complexity of real-time fuzzy control algorithms, especially as the volume of sensor data increases [22,29]. Furthermore, there is a lack of in-depth research on the automatic generation of fuzzy rules, which would minimize human involvement in control system tuning [30]. Advancing these areas will enhance the efficiency and effectiveness of cobot applications.

The main objective of this article is to review existing research and control methods for collaborative robots. It examines both classical control approaches and the fundamental principles of fuzzy logic to highlight its key components for solving control problems. Special attention is given to the advantages of fuzzy logic under uncertain conditions, its integration with other methods, and its potential to improve cobot adaptability through flexible control strategies.

2. Fundamentals of Fuzzy Logic and Its Application in Control

Collaborative robots require adaptive and flexible control systems since they interact directly with humans [31]. In this case, it is necessary to account for the degree of uncertainty in real time. Traditional control methods, such as PID control or mathematical behavior models, cannot provide the required level of adaptability to changes in human behavior and unpredictable environmental factors. Therefore, a more flexible control solution must accommodate these factors [32,33]. A fuzzy logic-based control system can provide an effective solution by replicating human reasoning and determining control trajectories based on fuzzy rules rather than precise mathematical calculations [34,35].

To assess the adaptability of a fuzzy logic-based control system, it is important to understand its key components: fuzzy (linguistic) variables and their membership functions; fuzzy rules and inference mechanisms; aggregation and defuzzification [34,36].

2.1. Fuzzy Variables and Membership Functions

Fuzzy variables in a fuzzy logic-based control system are linguistic variables commonly used in everyday life. For example, in a cobot control system, the following linguistic variables can be applied [37,38]:

- For distance evaluation—“close distance”, “medium distance”, “far distance”;
- For speed evaluation—“low speed”, “medium speed”, “high speed”;
- For assessing sudden factors such as obstacles—“obstacle present”, “no obstacle”, or “minor congestion”, “moderate congestion”, “significant congestion”.

Unlike binary logic, which has strict boundaries (true or false; 0 or 1), fuzzy logic indicates the degree of membership of a variable to a certain value. Each fuzzy set is defined by a membership function, which assigns a degree of belonging to a particular set of values. The most common membership functions are triangular (1), trapezoidal (2), and Gaussian (3) functions. The choice of function depends on the required type of control (smoother, more flexible, or stricter). Fuzzy logic membership functions allow cobots to

interpret sensor data similarly to human perception, enabling an adaptive control system in a dynamic environment [39,40].

$$\text{triangle}(x; a, b, c) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{c-a}, & a \leq x \leq c \\ \frac{b-x}{b-c}, & c \leq x \leq b \\ 0, & b \leq x \end{cases} \quad (1)$$

where a and b —borders of the triangle function; c —the center of the triangle function.

$$\text{trapezoid}(x; a, b, c, d) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{c-a}, & a \leq x \leq c \\ 1, & c \leq x \leq d \\ \frac{b-x}{b-d}, & d \leq x \leq b \\ 0, & x \leq b \end{cases} \quad (2)$$

where a and b —borders of the trapezoid function; c and d —the maximum of the trapezoid function.

$$\text{gaussian}(x; c, \sigma) = e^{-\frac{1}{2}(\frac{x-c}{\sigma})^2}, \quad (3)$$

where c —the middle of the gaussian function; σ —the width of the gaussian function.

2.2. Fuzzy Rules and Inference Mechanism

The core of a fuzzy logic-based control system is a set of fuzzy rules structured as “condition–conclusion” (if–then). In these rules, linguistic variables interpret real inputs and outputs, thereby defining control actions. Examples of fuzzy rules include [35,39]:

- If the distance to an object is “close distance,” then the cobot’s speed is set to “low speed”;
- If a sudden factor is classified as “slight congestion,” then the cobot’s speed is determined as “high speed”.

These rules are built based on empirical and expert data, allowing the development of an intuitive control system that makes decisions without requiring a precise mathematical model [38].

The fuzzy logic inference mechanism consists of two stages [38]:

1. Fuzzification—converting real input values (speed, distance, etc.) into fuzzy values using predefined membership functions.
2. Linguistic variable inference—combining fuzzy rules to determine the appropriate output based on input conditions, resulting in a fuzzy conclusion.

2.3. Aggregation and Defuzzification

Since a single fuzzy variable can belong to multiple sets with varying degrees of membership, applying fuzzy rules can generate multiple fuzzy outputs. The aggregation process combines these outputs into a single fuzzy decision. Different methods are used for output aggregation, such as the maximum/minimum method or the weighted average method. The selected method determines the control strategy, which is later adjusted based on experimental data [34,36].

The final step in converting a fuzzy decision into a numerical value is defuzzification. Several methods are used at this stage, including [34,36]:

1. Center-of-gravity method, which calculates the centroid of a fuzzy set to determine a precise value.

2. Weighted average method, which computes a numerical value based on the sum of all contributing fuzzy sets.

2.4. Advantages of Fuzzy Logic in Cobot Control

The adaptability and flexibility of fuzzy logic enable cobot control systems to adjust to various conditions and operate in different environments. The main advantages of fuzzy logic include [33,41,42]:

- Robustness to uncertainty, as fuzzy logic processes incomplete data, making it a useful tool in human–robot interaction scenarios;
- Simplification of decision-making, eliminating the need for a precise mathematical model and simplifying implementation and tuning;
- Smooth and adaptive control, ensuring gradual control adjustments and robot movements, reducing the risk of hazardous interactions;
- Enhanced safety, achieved through dynamic force regulation based on sensor data.

Thus, fuzzy logic-based control systems are widely applied for force control, trajectory planning, movement strategy planning, and real-time action adjustments. Additionally, integrating fuzzy logic with other methods (e.g., machine learning) enhances accuracy, autonomy, and adaptability in cobot control.

3. Collaborative Robots: Characteristics and Challenges

The evolution of industrial automation has led to the emergence of collaborative robots (cobots) as a transformative and pivotal technology, designed to work alongside human operators in shared workspaces, effectively bridging the gap between fully automated systems and manual labor [43–46]. Their design philosophy centers on enabling safe and efficient interaction with human workers in shared workspaces, offering a unique blend of robotic precision and human adaptability [47]. This section explores the distinctive characteristics of cobots and the significant challenges in their control systems, highlighting the need for advanced control methodologies such as fuzzy logic.

3.1. Features of Collaborative Robots

Unlike traditional industrial robots that operate in isolated environments, collaborative robots incorporate numerous safety and adaptability features that enable direct human–robot interaction. The defining features of cobots are intrinsically linked to their collaborative nature, emphasizing safety, seamless human interaction, industrial applicability, and the ability to adapt to dynamic environments [48,49]. However, the implementation and effectiveness of these features can vary significantly across different cobot designs and interaction capabilities tailored to different industrial needs. Below is a comparative analysis of notable cobot models, highlighting their safety, adaptability, human–robot communication interfaces, and overall effectiveness in industrial applications.

3.1.1. Safety Features

Traditional industrial robots are typically large, dangerous, and built for specific processes, often requiring physical barriers such as safety cages or light curtains to protect workers. In contrast, cobots are designed with inherently safe characteristics that mitigate potential hazards during collaborative operations [50].

Modern cobots are equipped with sophisticated safety systems that comply with international safety standard ISO/TS 15066:2016 [51], which outlines four types of key safety functionalities for cobots to address four distinct collaborative operation modes: safety-rated monitored stop, hand guiding, speed and separation monitoring, power and force limiting [51–53]:

- **Safety-Rated Monitored Stop:** This feature ensures that the robot stops safely upon human entry into the collaborative workspace, resuming only when the human leaves.
- **Hand Guiding:** Operators can manually guide the robot, enhancing control and safety during collaborative tasks. This feature is particularly useful for tasks requiring precision and human intervention.
- **Speed and Separation Monitoring:** The robot dynamically adjusts its speed based on the distance to the human worker, slowing down or stopping entirely if a predefined safety distance is breached. This dynamic adjustment helps in preventing collisions and ensuring safe operation.
- **Power and Force Limiting:** Cobots are equipped with force sensing capabilities that allow them to detect abnormal forces and stop their motion immediately to prevent injuries. This feature helps in reducing the impact of potential collisions and avoiding certain types of incidents, such as crushing accidents.

Each mode represents a different approach to ensuring human safety while maintaining operational efficiency. For instance, power and force limiting ensures that the robot cannot exceed a predetermined force threshold, while speed and separation monitoring maintains a safe distance between the human operator and the robot [50,52].

Cobots are designed to be lightweight, flexible, and easy to reprogram, which enhances their safety and usability in various applications. Ergonomic considerations ensure that cobots can be integrated into workspaces without causing strain or injury to human workers. To minimize injury upon contact, cobots often feature rounded edges and soft padding [54,55].

The ergonomic structures are complemented by advanced sensing capabilities that constantly monitor the working environment with real-time control updates to prevent accidents or injuries during collaborative tasks. A range of hardware and software features, such as motion detection, protective stops, safety-rated sensors, collision detection are commonly integrated into cobots to contribute to the overall safety of the collaborative operation and prevent accidents and injuries [53].

All cobots prioritize safety, but the methods employed can differ, Table 1 presents a comparative analysis of notable cobot models (Universal Robots' UR Series, ABB's YuMi, KUKA's LBR iiwa, FANUC's CR Series), emphasizing safety features.

Table 1. Safety feature comparison.

Model	Key Safety Features	Industrial Suitability
Universal Robots UR Series	Power & force limiting, software safety functions, speed & separation monitoring, hand guiding [53,56].	Broad range of applications (palletizing, tending, assembly, etc.), good for SMEs (small and medium-sized enterprises) and flexible production [53,56].
ABB YuMi	Inherent safety (lightweight, rounded, padded), software collision detection.	Primarily for small parts assembly, especially in electronics, requiring high precision and close human interaction [57,58].
KUKA LBR iiwa	Highly sensitive joint torque sensors, certified safety functions, force control [53,59,60].	Tasks requiring high sensitivity and precision (assembly, machining, handling, inspection), direct human collaboration [53,59,60].
FANUC CR Series	Safe contact stop sensors, customizable speed & safety settings, FANUC Hand Guidance [53,59,61].	Wide range of payloads for diverse tasks (assembly, pick & place, heavy lifting), suitable for various industries [53,59,61].

The comparison of the specified cobot series highlights distinct approaches to safety and varying degrees of industrial suitability that cater to different needs in collaborative robotics. Each series effectively addresses the fundamental safety requirements for human–robot interaction, albeit through different technological implementations, ranging from

inherent physical limitations to sophisticated sensor-based control and robust contact detection. In terms of industrial suitability, the UR Series stands out for its versatility across a broad spectrum of applications and its user-friendly nature, making it accessible to a wide range of industries and particularly beneficial for flexible manufacturing environments. ABB's YuMi is specifically engineered for delicate and precise small-part assembly tasks requiring close human collaboration. KUKA's LBR iiwa excels in applications demanding high sensitivity, accurate force control, and seamless interaction in complex manipulation scenarios. Finally, FANUC's CR Series offers a robust solution capable of handling a wide range of payloads, including heavier industrial materials, making it suitable for both traditional cobot tasks and more demanding material handling applications.

The diversity in safety mechanisms and industrial capabilities underscores the ongoing evolution and specialization within the field of collaborative robotics. The selection of an appropriate cobot necessitates a thorough understanding of the specific task requirements, the nature of the human–robot interaction, and the environmental constraints of the intended application. As technology continues to mature, future advancements will likely focus on further enhancing the safety, adaptability, and ease of integration of cobots across an even wider range of industrial sectors.

3.1.2. Human–Robot Interaction

The fundamental characteristic that distinguishes cobots from conventional industrial robots is their ability to interact directly with human operators in a shared workspace. This interaction can take various forms, from simple co-existence to complex collaborative tasks. In complex collaboration scenarios, shared task planning and synchronized execution become essential. The design philosophy behind cobots emphasizes intuitive interfaces that facilitate natural interaction, making them accessible to operators with minimal specialized training [62].

For effective human–robot collaboration, the role of the robot should transition naturally and smoothly between that of a leader and a follower to guarantee optimal task performance. This dynamic role distribution requires sophisticated control architectures that can interpret human intentions and respond appropriately. The arbitration of shared control between the human and the robot requires well-defined strategies within the control architecture to ensure efficient task execution and safety [48,63].

Effective collaboration requires more than just physical safety; it demands intuitive and productive interaction between humans and robots. Key aspects of human interaction in cobotics include the following:

- **Intuitive Programming and Operation:** Cobots often feature user-friendly interfaces, such as hand guiding or graphical programming, making them accessible to users without extensive robotics expertise [64,65].
- **Ergonomics and Task Sharing:** Cobots are intended to assist humans with physically demanding, repetitive, or potentially hazardous tasks, improving workplace ergonomics and reducing the risk of injury [64].
- **Communication and Trust:** While still evolving, future cobots may incorporate more sophisticated communication methods to enhance understanding and trust between human and robot workers. Predictable and safe robot behavior is crucial for building operator confidence [46,49,66].

Table 2 presents a comparative analysis of human–robot interaction (HRI) aspects for leading cobots models, focusing on programming, ergonomics, and trust-building features.

Table 2. HRI aspect comparison [56,58,60,61].

Model	Ease of Programming	Ergonomics & Task Sharing	Communication & Trust
Universal Robots UR Series	Intuitive graphical user interface (PolyScope) with drag-and-drop functionality, easy hand guiding (teach pendant). Focus on user-friendliness for non-expert programmers. Offline programming options.	Designed to assist with ergonomically challenging and repetitive tasks, reducing strain on human workers. Lightweight design and flexible deployment facilitate easy integration into dynamic workflows. 360° mounting for workspace flexibility.	Visual cues through the teach pendant interface provide real-time information about the robot's state and planned actions. Focus on predictable and consistent robot behavior to foster operator confidence.
ABB YuMi	Intuitive lead-through programming via hand guidance on the robot arms. Simplified interface designed for quick task setup. Wizard-based programming tools and offline programming capabilities within RobotStudio. Emphasis on assembly tasks.	Specifically designed for small parts assembly, improving ergonomics for intricate and repetitive manual tasks. Dual-arm configurations allow for complex task sharing, mimicking human bimanual manipulation. Focus on collaborative work on assembly lines.	Physical co-presence and design (dual arms mimicking human interaction) to create a more intuitive collaborative experience. Emphasis on safe and predictable movements to build operator trust, particularly in close-proximity assembly tasks.
KUKA LBR iiwa	Teaching by manual guidance with high sensitivity due to torque sensors, allowing for precise trajectory definition. KUKA Sunrise. OS programming environment, offering both graphical and text-based options.	Torque-sensitive joints for safe, direct collaboration in tasks requiring delicate manipulation and precise force application. Dynamic payload calibration.	AR visualization of task progress and intent. Highly sensitive torque sensors allow the robot to react naturally to human contact, potentially fostering a sense of safety and trust.
FANUC CR Series	FANUC Hand Guidance for intuitive teaching by physically guiding the robot. iHMI (Intelligent Human-Machine Interface) with simplified screens and easy-to-understand icons. Offline programming with ROBOGUIDE. User-friendly programming wizards.	Wide range of payload capacities allows for handling both light and heavy tasks. Focus on tasks like material handling, assembly, and machine tending where the robot can take over physically demanding or repetitive actions. Safe task sharing via pre-defined paths. Push-back function for safety.	FANUC Hand Guidance provides direct physical interaction for teaching and intervention. Clear status indicators. Focus on ensuring predictable and safe responses during collaborative tasks to build trust in the robot's capabilities. Lacks real-time intent communication

The comparative analysis reveals that while all four collaborative robot series prioritize safe and effective human–robot interaction, they exhibit distinct approaches and strengths in their HRI design. Universal Robots emphasizes ease of programming through intuitive interfaces and hand-guiding, alongside ongoing research into advanced programming methods, making them accessible to a broad range of users. Their focus on predictable behavior contributes to building operator trust. ABB's YuMi, designed for close-proximity assembly, leverages its dual-arm design and intuitive lead-through programming to facilitate seamless interaction in shared workspaces. KUKA's LBR iiwa employs highly sensitive torque sensors, enabling natural and responsive interaction, with research exploring advanced communication through motion and force feedback. FANUC's CR Series utilizes hand guidance and clear status indicators, with a focus on robust safety features and predictable responses to enhance operator confidence, while research explores more sophisticated communication methods. These differences highlight the diverse strategies employed by manufacturers to address the complexities of HRI. Universal Robots aims for user-friendly versatility, ABB focuses on intuitive collaboration in assembly, KUKA emphasizes sensitive and nuanced interaction, and FANUC balances ease of use with robust safety for industrial applications. The ongoing research into advanced programming, augmented reality, and more expressive robot behaviors indicates a continued commitment to improving HRI, with the goal of achieving safer, more intuitive, and more productive human–robot collaboration in industrial settings.

Human–robot collaboration presents unique challenges related to the unpredictable nature of human behavior. The working environment of cobots is subjected to unforeseeable modifications caused by people, requiring control systems that can adapt to these changes without compromising safety or efficiency [67]. This necessitates a delicate balance between robot autonomy and human control, where neither dominates the interaction entirely.

3.1.3. Adaptability and Flexibility

Cobots are frequently deployed in dynamic manufacturing environments where tasks and workflows can change rapidly and demand adaptability to different working processes. Unlike traditional industrial robots that are programmed for specific repetitive tasks, cobots can be quickly reprogrammed and redeployed for various functions. This flexibility allows them to be used in various industries and applications, from manufacturing to healthcare [48].

Key characteristics of adaptability include the following [48]:

- **Flexibility in Task Execution:** Cobots can be easily reprogrammed and redeployed for a variety of tasks, often with minimal downtime.
- **Environmental Awareness:** Utilizing various sensors like vision systems, force/torque sensors, and proximity sensors, cobots can perceive and react to changes in their environment, such as the presence of humans or obstacles.
- **Robustness to Uncertainty:** Cobots need to operate reliably in the presence of real-world uncertainties, including variations in object placement and human movements.

The adaptability of cobots is enhanced through the integration of advanced sensing technologies, particularly vision systems that enable them to perceive and respond to changes in their environment.

Table 3 presents a comparative analysis of adaptability and flexibility features of the selected cobots models, focusing on their capabilities in task execution, environmental awareness, and uncertainty handling.

Table 3. Adaptability and flexibility feature comparison [56,58,60,61].

Model	Flexibility in Task Execution	Environmental Awareness	Robustness to Uncertainty
Universal Robots UR Series	Easy reprogramming for different tasks. Wide range of accessories and applications for enhancing task versatility. AI-driven path planning for high-speed pick-and-place and palletizing. AI Accelerator for dynamic task switching (e.g., machine tending to quality inspection).	Compatible with various external sensors (vision-guided systems with real-time object detection) for enhanced environmental perception. Built-in safety functions (AI-powered obstacle avoidance) contribute to safe operation in dynamic environments with human presence.	Self-correcting trajectories for unexpected object shifts. Adaptive force control in machining tasks. Scripting and programming flexibility allow for developing adaptive routines to handle some uncertainties. Relies on programmer to account for variability.
ABB YuMi	Designed for agile production and small parts assembly, easily adaptable to new products or assembly sequences. Two arms for complex manipulation and task sharing. Lightweight design and simplified programming for rapid changeovers between tasks.	Some models have integrated vision systems for part location and inspection. Built-in force and torque sensing provides awareness of contact forces during interaction. Collision detection via Hall-effect sensors.	Dual-arm redundancy for error recovery in assembly tasks. Software algorithms help compensate for minor variations in part placement. Real-time collision pause. Designed for structured tasks with limited variability.
KUKA LBR iiwa	High degree of freedom (7 axes) for human-like arm flexibility and adaptation to complex workspaces. Wide range of programming options, software tools and applications. Various control modes for different task requirements (e.g., assembly, force-sensitive tasks). AR-guided task mapping for visual reprogramming.	Joint torque sensors in all axes for proactive collision avoidance. Interfaces for integrating external sensors (e.g., vision, external force/torque) for comprehensive environmental perception. Advanced control for compliant motion in uncertain environments. Mobile platform compatibility.	Advanced control strategies like impedance control make the robot robust to external disturbances and uncertainties during interaction. Real-time sensor feedback to handle unforeseen situations. High level of adaptation to dynamic changes.
FANUC CR Series	Adapting to various tasks through different configurations and end-effector options. Simply reprogrammable for new tasks. Compatibility with other FANUC solutions (e.g., vision systems, mobile robots).	Compatible with various sensors, including FANUC iRVision and force sensors, to perceive and respond to changes in the workspace. Safe contact stop and speed/separation monitoring contribute to safe operation in shared environments. Integration with external devices.	Force sensing capabilities allow the robot to adapt to variations in contact forces during tasks like assembly or grinding. Features and programming options to accommodate some level of uncertainty in the environment. Integration with vision systems for error recovery.

The current comparative analysis shows that collaborative robot manufacturers are using different strategies to achieve adaptability and flexibility, which are key characteristics for effective deployment in dynamic industrial environments. Universal Robots prioritizes ease of reprogramming and a broad ecosystem of compatible components, facilitating flexible task execution across diverse applications. ABB's YuMi, while designed for a more specific domain of small parts assembly, offers flexibility through its dual-arm dexterity and quick changeover capabilities within that domain. KUKA's LBR iiwa leverages its high degree of freedom and advanced control modes, particularly impedance control, to excel in scenarios demanding robustness to uncertainty and fine-grained adaptation to complex tasks. FANUC's CR Series emphasizes versatile application support and integration with FANUC's wider automation ecosystem, providing flexibility across a range of payloads and industrial settings. The emphasis on environmental awareness also varies. While all series can integrate sensors, KUKA's LBR iiwa, with its inherent joint torque sensors, demonstrates a greater capacity for real-time, force-based adaptation. ABB's YuMi uses integrated vision for structured environments, and FANUC and Universal Robots support a range of sensor options for broader awareness. These differences highlight the ongoing development of collaborative robots toward meeting diverse and evolving industrial needs.

Many collaborative robots employ AI algorithms and optical sensors to navigate industrial environments without constant human guidance. For instance, Universal Robots' UR10e cobots utilize sensor fusion—including LiDAR, ultrasonic, and pressure sensors—to dynamically map their surroundings and adjust movements in real time, ensuring safe operation alongside human workers in busy production lines. Similarly, ABB's cobots implement virtual safety zones with cameras and sensors that automatically slow or stop the robot when a human enters a defined area, enabling seamless human–robot collaboration without physical barriers [56,58].

However, these vision-based systems often depend on prior knowledge of the objects to be manipulated, limiting adaptability in highly dynamic or unstructured environments. To overcome this, recent research and industrial implementations are increasingly incorporating machine learning techniques, especially reinforcement learning (RL), to enable cobots to continuously learn and adapt to unforeseen changes. For example, in automotive assembly, dual-arm collaborative robots equipped with AI-driven control systems have demonstrated improved productivity and flexibility by adapting to variations in part positioning and handling large components safely alongside human operators. In logistics, sensorized robotic systems with machine learning capabilities manage bulky and diverse objects, adjusting grasping and manipulation strategies on the fly while maintaining worker safety. Reinforcement learning approaches have also been explored for pick-and-place tasks, allowing cobots to grasp previously unseen objects by learning optimal grasping policies through trial and error in simulation before deployment in real environments [68,69].

Similarly, fuzzy logic has been applied to improve human–robot interaction by interpreting ambiguous or imprecise human commands, enabling robots to adjust their actions dynamically. For instance, fuzzy logic can be used in mid-level controllers for wearable assistive devices and rehabilitation robots, providing the necessary torque adjustments based on the robot's dynamics and external loads. This approach helps in simulating realistic human behavior and improving the interaction experience [70].

These developments illustrate a significant research trend combining machine learning and fuzzy logic to enhance cobot adaptability. Reinforcement learning provides the capability for continuous environmental learning and task generalization, while fuzzy logic offers robust decision-making under uncertainty and imprecision. Together, they represent a promising frontier for developing collaborative robots capable of flexible, reliable operation in complex, dynamic industrial settings.

3.2. Control System Requirements

Traditional industrial robot control methods, such as Proportional-Integral-Derivative (PID) control, trajectory tracking, force control, often focused on precision and repeatability in structured environments, is insufficient for the complexities of human–robot collaboration. The control systems of cobots requires ensure both efficient and safe operation [71].

Based on international robotic safety standards, industry best practices, and recent literature, the requirements are identified and ranked by significance in Table 4.

Table 4. The significance of the requirements [67,68,72,73].

Requirement	Description	Importance	Rationale
Safety Assurance	Systems to ensure safe operation around humans, including emergency stops and collision detection. The safety functionalities described in ISO/TS 15066.	Very High	Critical for preventing accidents and ensuring user safety.
Real-time Performance	Ability to respond to human commands and environmental changes in real-time.	High	Critical for natural interaction and avoiding collisions; addresses unpredictable human behavior in shared spaces.
Adaptability and Flexibility	Capability to reprogram quickly and handle multiple tasks or payloads	Medium–High	Essential for effective interaction and collaboration with humans.
Robustness and Stability	Ensures reliable operation in the presence of uncertainties, disturbances, and variations in human behavior.	Medium	Key for maximizing the return on investment and meeting evolving manufacturing needs.
Fault Tolerance and Reliability	Minimizes downtime and ensures operational continuity in industrial settings.	Medium	While less directly related to immediate safety, failures can disrupt production and lead to economic losses
Integration of Multiple Sensors	Allows cobots to perceive their environment more accurately and respond appropriately to complex situations.	Medium	Important for advanced collaboration and handling a wider range of tasks.

Cobots predict human intentions and adapt their behavior accordingly for natural and efficient collaboration. This involves incorporating models of human motion and decision-making processes using data sources like force/torque measurements, visual tracking, and historical interaction patterns. Robust sensor fusion algorithms are needed to operate in real-time while accommodating sensor noise and uncertainty. Control systems must balance multiple objectives, including task completion, safety, efficiency, and user comfort [48,74].

Cobots achieve precise motion synchronization with humans, even during tasks involving physical contact or shared manipulation of objects. Control systems must implement robust disturbance rejection algorithms to distinguish between intentional guidance and unintended perturbations. Fuzzy logic-based approaches can continuously regulate control authority based on interaction forces, system velocity, and environmental factors, enabling smooth transitions between collaborative modes [48,75].

Controllers for human–robot collaboration must consider adjustable autonomy and mixed initiative. Adjustable autonomy allows the dynamic allocation of control authority between the human and the robot based on task context and requirements. Mixed initiative enables both the human and the robot to initiate actions and make decisions within the collaborative framework. Control systems must balance autonomy and initiative to achieve optimal collaboration, considering both technical capabilities and human factors like user experience and trust [46,49,66,68].

Based on the literature review, collaborative robots represent a significant paradigm shift in industrial automation, offering unprecedented opportunities for human–robot collaboration in shared workspaces. Their distinctive features—safety-oriented design, natural human interaction capabilities, and adaptability to diverse workflows—enable applications that were previously infeasible with traditional industrial robots. However, these features also introduce complex control challenges that require innovative solutions.

Control system requirements for cobots encompass safety architectures, human intention prediction, motion synchronization, and the delicate balance between autonomy and initiative. Addressing these requirements demands control methodologies that can handle uncertainty, adapt to changing conditions, and facilitate natural human–robot interaction. Fuzzy logic, with its ability to process imprecise information and model human-like reasoning, represents a promising approach for meeting these challenges and realizing the full potential of collaborative robotics in industrial applications.

Building on this understanding of cobot characteristics and control challenges, the next section will explore specific applications of fuzzy logic in collaborative robot control, including control architectures, hybrid approaches, and techniques for enhancing adaptability, prediction, and motion correction.

4. Fuzzy Logic for Cobot Control

This chapter synthesizes recent studies focusing on fuzzy control architectures, hybrid approaches integrating fuzzy logic with other methods, and innovations aimed at augmenting the adaptability and precision of robotic movements.

4.1. Control Architectures

Fuzzy inference systems (FIS) have become indispensable tools in decision-making and control applications across various domains. Among the most prevalent models are the Mamdani fuzzy inference model and the Sugeno fuzzy inference model, each possessing its unique characteristics and applications. This section explores the distinctions, implementations, and practical implications of these two models, supporting insights with recent scholarly research [76,77].

4.1.1. Structure and Mechanism of Mamdani Fuzzy Model

The Mamdani fuzzy model, introduced by Ebrahim Mamdani in 1975 for controlling a steam engine, remains prominent due to its intuitive approach towards handling uncertainty and imprecision in systems [78]. The general structure of the Mamdani model is defined as follows:

1. **Fuzzification:** Input variables are converted into fuzzy sets using predefined membership functions. Typically, this step involves linguistic variables, such as “high,” “medium,” or “low,” which can be represented by various shapes (triangular, trapezoidal) [79].
2. **Rule Evaluation:** Fuzzy rules in the antecedent typically follow the format:
3. IF x_1 is A_1 AND x_2 is A_2 THEN y is B .
4. where A_1 and A_2 represent fuzzy membership functions, and B is the output fuzzy set. The evaluation of these rules involves the fuzzy operators AND, OR, and NOT, which are used to derive a fuzzy conclusion for each rule [80].
5. **Aggregation:** The outputs of all rules are combined to create a single fuzzy set that represents all output conditions.
6. **Defuzzification:** The final step involves converting the aggregated fuzzy set into a crisp output, often using techniques like the centroid method, which calculates the center of area under the curve of the fuzzy set [81].

Mamdani systems have been utilized in diverse applications, including temperature control, where (Jabeur and Seddik Jabeur & Seddik [82]) showcased its implementation for mobile robot navigation, yielding robust performance against unexpected obstacles.

4.1.2. Sugeno Fuzzy Model

The Sugeno fuzzy model, also known as the Takagi-Sugeno (T-S) model, simplifies the inference process and often serves better in control applications [83]. Sugeno models are characterized by:

1. Fuzzification: Similar to the Mamdani model, inputs are fuzzified into fuzzy sets using membership functions [79].
2. Rule Evaluation: The rules take on a different form where the conclusion is expressed as a mathematical function:
3. IF x_1 is A_1 AND x_2 is A_2 THEN $y = f(x_1, x_2)$
4. where f is typically a linear or constant function. This avoids the complexity of calculating fuzzy outputs by returning crisp results directly from the rules [84].
5. Aggregation and Defuzzification: In Sugeno systems, since the output is already a number, the defuzzification is performed through a weighted average of the outputs of each rule, providing a more straightforward computation process compared to the Mamdani approach [85].

Research indicates that Sugeno models often result in higher computational efficiency and speed, especially in real-time applications. For instance, in context of diagnosing diseases Sugeno's efficiency relative to the Mamdani approach, achieving an error rate lower than the Mamdani method [86,87].

4.1.3. Performance Comparison and Application Domains

The choice between Mamdani and Sugeno models can significantly impact the performance of fuzzy inference systems in various applications. Quantitative comparisons emphasize these differences, particularly in control scenarios.

1. Control Systems: Sugeno models are often preferred in control systems requiring rapid responses. For example, in a comparative study of fuzzy logic controllers for photovoltaic systems, both Mamdani and Sugeno controllers were implemented, and the Sugeno controller had a faster response time and less output variation during transient states [88].
2. Predictive Models: Sugeno methods are recognized for their forecasting accuracy and are widely deployed in predictive scenarios due to their ability to approximate complex functions efficiently. Study reports [89], that the Sugeno approach achieved a prediction accuracy significantly outperforming the Mamdani model in a similar context.
3. Interpretability and Complexity: The Mamdani model offers greater interpretability through elaboration on fuzzy sets, which many users find beneficial. Researchers have noted that while the Sugeno model tends to produce better computational efficiency, the linguistic clarity present in Mamdani's model aids in explaining system behaviors to non-specialists [90].

Both models have their advantages in different contexts, and this has been observed in agricultural applications. For instance, utilizing both models to predict temperature conditions in poultry farming, the Sugeno model surpassed Mamdani in execution time but required additional effort in presenting the results meaningfully [91].

In mobile robotics, comparative study [92] revealed that a Sugeno-based fuzzy logic controller outperformed Mamdani in line-following tasks due to better computa-

tional efficiency. Notably, it reduced the deviation from the target track significantly in dynamic environments.

In another investigation of household light control systems [93], the implementation of a Sugeno fuzzy model resulted in decreased energy consumption compared to a Mamdani model by optimizing the light intensity supply based on environmental changes through faster decision-making.

In summary, while both Mamdani and Sugeno fuzzy models have their unique strengths, the choice between them often relies on the specific application and requirements of system performance. Mamdani models remain desirable for tasks requiring interpretability and linguistic output, while Sugeno models prevail in applications demanding rapid calculations and precise numerical outputs.

Fuzzy logic has proliferated various architectures designed to address complex decision-making and control challenges in a multitude of fields. While the Mamdani and Sugeno models remain prominent, other fuzzy logic architectures have emerged, showcasing unique strengths and capabilities. In the next subsections we explore several alternative fuzzy logic architectures and compare them with the Mamdani and Sugeno models.

4.1.4. Fuzzy Cognitive Maps (FCMs)

Fuzzy Cognitive Maps (FCMs) combine fuzzy logic with graph theory, representing knowledge as a directed graph where weighted edges indicate the influence of one concept on another. This structure makes FCMs particularly well-suited for modeling complex systems in which relationships and dependencies between components are critical. FCMs have been explored in the context of energy management systems, effectively capturing interactions among multiple energy sources and enabling optimization of their contributions in response to fluctuating demand [94].

FCMs also offer advantages in scenario analysis and decision support, allowing practitioners to simulate the effects of varying inputs or interventions on the overall system. In addition, the ability to dynamically update edge weights based on system feedback makes FCMs adaptable to changing conditions, enhancing their relevance for real-time decision-making and strategic planning. Future research may explore hybrid approaches that integrate FCMs with quantitative fuzzy systems or machine learning techniques, combining interpretability with numerical precision to address complex control and optimization challenges [94].

Unlike the Mamdani and Sugeno models, FCMs do not rely on crisp outputs and are more qualitative in nature, which can lead to more intuitive insights into system behavior. However, the discussion on precision relative to Mamdani and Sugeno may require additional nuance, as the trade-offs between qualitative modeling and numerical outputs can vary contextually.

4.1.5. Adaptive Neuro-Fuzzy Inference Systems (ANFIS)

Adaptive Neuro-Fuzzy Inference Systems (ANFIS) merge artificial neural networks with fuzzy logic, enabling dynamic adaptation of fuzzy rules based on input data. This hybrid approach allows ANFIS to combine the learning capabilities of neural networks with the interpretability of fuzzy logic, providing a flexible framework for both function approximation and data-driven decision-making. One of the key strengths of ANFIS is its ability to model nonlinear systems and adjust its inference rules in response to changing conditions, making it particularly suitable for complex, real-time applications [95].

Despite its potential, the exact performance improvements of ANFIS compared to traditional Mamdani systems are not always well quantified in the literature, highlighting the need for more systematic benchmarking studies. Nevertheless, practical applications

demonstrate its versatility. ANFIS has been successfully implemented in a variety of real-time control scenarios, including automotive systems, robotics, and process control, where rapid adaptation to varying inputs is crucial. For example, in automotive control, ANFIS can optimize engine performance or vehicle stability by continuously tuning control parameters in response to environmental changes. Similarly, in robotics, ANFIS enables adaptive trajectory planning and motion control, improving precision and responsiveness in dynamic environments [95].

Overall, ANFIS represents a promising avenue for systems that require both adaptive learning and interpretable control strategies, bridging the gap between data-driven modeling and rule-based reasoning. Future research could focus on hybridizing ANFIS with emerging AI techniques, such as reinforcement learning or deep learning, to further enhance adaptability and performance in increasingly complex control tasks.

4.1.6. Comparative Analysis

The comparison of fuzzy logic architectures reveals key differences in application suitability and performance metrics:

- Mamdani vs. Sugeno: Mamdani is ideal for user-friendly systems requiring linguistic interpretability, while Sugeno is preferred in applications demanding efficiency and faster computations.
- FCM vs. Traditional Models: FCMs excel in illustrating complex interdependencies, but precise numeric outputs might be necessary for quantitative control applications, depending on use cases.
- ANFIS vs. Static Fuzzy Models: ANFIS adapts dynamically to changes in input data, which is beneficial for systems that encounter variable conditions in real time.

As the field of fuzzy logic continues to evolve, these various architectures demonstrate the flexibility and adaptability of fuzzy systems to different problem domains. While Mamdani and Sugeno models remain foundational, alternative architectures like FCMs and ANFIS offer significant advances for managing uncertainty in decision-making and control systems.

4.2. Hybrid Control Architectures

The advancement of collaborative robot (cobot) control systems using hybrid approaches involving fuzzy logic represents a significant stride in the field of robotics. The integration of fuzzy logic with other intelligent systems—particularly neural networks, genetic algorithms, and adaptive control techniques—allows for greater flexibility, robustness, and efficiency in navigating complex environments. This synthesis focuses on various hybrid approaches and their applications in cobot control, drawing from recent literature to elucidate the state of the art in this domain [96].

The emergence of hybrid systems that combine fuzzy logic with other methodologies further illustrates the potential for enhanced control architectures. A model integrating fuzzy logic and neural networks, showing its capacity to navigate the complexities of dynamic environments [96]. Such hybrid fuzzy logic systems provides the necessary flexibility, which allows for adaptive tuning of parameters based on environmental feedback, demonstrating a marked improvement in trajectory accuracy. This hybrid system provided up to 30% improved performance over conventional methods when tested in dynamic scenarios, underscoring the efficacy of combining fuzzy architectures with machine learning techniques [96–98].

Hybrid control approaches that integrate fuzzy logic with other control methodologies, such as sliding mode control and neural networks, have garnered attention for their ability to enhance stability and adaptability in robot navigation. A Type-2 Fuzzy-Sliding Mode

Controller was developed [97] that effectively navigates mobile robots in environments with dynamic targets. Their simulations indicated that this hybrid control system reduced tracking errors by over 40% compared to traditional sliding mode controllers [97,98].

Furthermore, the application of a fuzzy logic framework for obstacle avoidance in differential-drive mobile robots was demonstrated [98]. The implementation demonstrated a significant reduction in collision rates, achieving an accuracy of 92% in avoiding obstacles compared to a baseline controller, which had a mere 65% success rate. This improvement highlights the effectiveness of fuzzy logic in handling unpredictable obstacles in real time [98].

A significant contribution to this field is the use of fuzzy logic in conjunction with genetic algorithms for optimization. Employing a hybrid fuzzy-genetic approach for path planning in dynamic environments, achieved path lengths that were notably shorter than those derived from conventional algorithms, demonstrating how evolutionary strategies refine fuzzy logic performance [99].

Lastly, the combination of fuzzy logic with interval type-2 systems offers robustness against uncertainties. For example, evaluating its effectiveness in navigating agricultural robots, revealed improvements in decision-making accuracy in comparison to traditional fuzzy methods [100].

Hybrid fuzzy logic control systems commonly combine classical fuzzy logic with neural networks, where the latter serves to enhance the adaptability of robotic systems in uncertain and dynamic environments. For instance, the implementation of a hybrid neuro-fuzzy approach for a mobile robot designed for monitoring wall structures was explored [101]. The findings indicate that this integration not only allows for improved behavior control but also affords the robot the ability to learn from its environment, adapting its actions through experience. Similarly, another study [102] utilizes a fuzzy logic system alongside neural networks and disturbance observers to create a robust control methodology for a robot manipulator. The results demonstrated improved handling of disturbances and uncertainties that typically affect traditional control systems.

Moreover, the role of genetic algorithms in refining fuzzy controllers is evident in various studies. A genetic-fuzzy approach for optimizing trajectory planning and motion control in non-holonomic mobile robots navigating dynamic settings was presented [103]. This work highlights the effectiveness of hybrid solutions in enhancing the performance of fuzzy logic-based systems by systematically refining rule bases and membership functions. Furthermore, applying particle swarm optimization (PSO) to optimize the parameters of fuzzy logic controllers for mobile robot navigation, illustrates that optimal configurations can lead to efficient obstacle avoidance in complex environments [104].

In the context of leader-follower models, an adaptive fuzzy logic-based controller for wheeled mobile robots facilitating cooperative navigation was implemented [105]. The results underscored the improvement in maintaining formation during dynamic circumstances, showcasing the benefits of employing fuzzy logic in conjunction with adaptive techniques that respond to real-time changes in the operating environment. This highlights a fundamental characteristic of hybrid systems: their ability to leverage fuzzy logic's interpretability while augmenting adaptability with complementary methods.

In terms of application to path planning and obstacle avoidance, several hybrid approaches have demonstrated substantial improvements over conventional methods. For example, the implementation of hybrid fuzzy and adaptive control methods allows for real-time adjustments based on environmental feedback. An integrated approach was proposed [106] that combines fuzzy logic with an optimized artificial potential field (APF) method for enhanced trajectory planning and obstacle evasion. Their findings indicated

significant improvements in navigation efficiency, reinforcing the idea that hybrid systems can effectively operate even in unpredictable contexts.

Regarding the implementation of fuzzy logic in complex tasks, combining it with classical control techniques has also shown promise. A control system was developed that utilizes fuzzy logic alongside PID control for mobile robot navigation, demonstrating how hybrid configurations can capture the strengths of both systems. The results indicated superior performance in target tracking and obstacle evasion, thus confirming the effectiveness of such hybrid methodologies in improving control outcomes in robotic systems [82].

Another relevant approach is the use of fuzzy logic combined with simultaneous localization and mapping (SLAM) techniques. Collaboration between robots equipped with fuzzy logic controllers and advanced sensors can lead to improved decision-making capabilities in unstructured environments allowing robots to make informed decisions about their movement paths [96].

Furthermore, the employment of distributed architectures, where multiple robots operate collaboratively using fuzzy logic controllers, shows heightened robustness and scalability. For example, combining fuzzy logic and PID control for a group of two-wheeled balancing robots could achieve stability even under varying environmental conditions. This example emphasizes the adaptability fostered by hybrid approaches but also underscores their practical applications in collaborative scenarios [107].

The development of intelligent cleaning robots further exemplifies the power of fuzzy logic when integrated with AI techniques. By detailing a cleaning robot's autonomous obstacle avoidance through fuzzy logic-driven decisions based on sensory data. This highlights the potential benefits of integrating fuzzy logic with various AI paradigms to create more effective automated systems in domestic applications [108].

In conclusion, the application of hybrid approaches involving fuzzy logic in robot control systems facilitate enhancements in adaptability, robustness, and functionality in dynamic and complex environments. Through the fusion of fuzzy logic with neural networks, genetic algorithms, and adaptive control methodologies, research indicates improved performance in various applications, from simple navigation tasks to complex cooperative operations. The ongoing exploration of these hybrid methodologies promises to revolutionize the landscape of robotic control systems, providing avenues for greater efficiency and productivity in various sectors.

4.3. Enhancing Adaptivity, Prediction, and Motion Correction

Fuzzy logic has evolved significantly to provide powerful tools for enhancing the adaptability, predictability, and precision of robotic movements in various applications including collaborative robots (cobots), autonomous vehicles, and industrial automation. This section provides a detailed overview of fuzzy logic adaptations, technical details on architectures and methodologies, as well as comparisons of recent developments.

4.3.1. Enhancing Adaptivity with Fuzzy Logic Architectures

Adaptivity in robotic systems enables them to adjust their operations based on real-time data from their environment. One key approach is the implementation of fuzzy controllers, particularly hybrid fuzzy-PID controllers that combine the robustness of fuzzy logic with the stability of traditional PID control.

Such systems employ fuzzy logic to dynamically adjust PID controller parameters based on assessed error margins and system feedback, thereby enhancing responsiveness to environmental and operational changes. A hybrid differential evolution particle swarm optimization optimized fuzzy PID controller achieved a 25% reduction in the settling time and overshoot of power systems compared to standard PID controllers alone [109].

By applying an adaptive fuzzy-PID controller to control hydraulic systems, it was found that the hybrid model achieved a reduction in control errors from 7% to 2.5%, demonstrating significant improvements in tracking accuracy during varied load conditions [110].

An example of real-time adaptation can be seen in hydraulic actuation systems where fuzzy logic precisely adjusts parameters. For instance, using fuzzy control with adaptive learning algorithms, hyperbolic tangent models were utilized to provide adaptive control over suspension systems in automotive applications. That resulted in a decrease in body oscillations by approximately 30% compared to fixed-parameter controllers [111].

4.3.2. Prediction Capabilities in Fuzzy Logic Systems

Fuzzy logic's predictive capabilities play a crucial role in autonomous systems where anticipating future states is essential. By employing hybrid fuzzy inference systems that utilize predictive algorithms, cobots can optimize their strategies for navigation, task execution, and resource allocation.

A fuzzy-based hybrid decision-making algorithm in tri-rotor UAV stabilization was implemented [112]. The findings indicated a 45% reduction in error margins during dynamic flight conditions compared to traditional offset compensatory methods. The predictive model also incorporated historical data to dynamically adjust control inputs, forecasting external disturbances caused by varying weather conditions [112].

An adaptive path planning system was introduced [113] to autonomously guide vehicles in congested urban environments. Using a hybrid approach that combines adaptive fuzzy inference with predictive algorithms, a 38% increase in navigation efficiency was reported. The system could adjust trajectory in real-time by analyzing traffic density and road conditions, reducing the average time spent in traffic by nearly 10% [113].

4.3.3. Motion Correction Using Fuzzy Logic Techniques

Motion correction is essential for cobots, particularly when precise movements are needed in dynamic environments. Fuzzy logic aids in optimizing motion paths and compensating for deviations through adaptive control approaches.

A hybrid fuzzy controller for nonlinear half-vehicle suspension systems was developed [114]. This approach not only improved ride comfort but also achieved a noticeable decrease of 47% in response time for target trajectory following when compared to conventional control methods. They used fuzzy rules with variable membership functions to calculate the necessary adjustments based on real-time sensory feedback, enabling accurate motion correction [114].

Additionally, integrating hybrid fuzzy controllers with simulated annealing algorithms allowed adaptive path optimization. For example, a Takagi-Sugeno fuzzy model coupled with simulated annealing achieved a trajectory success rate of 90%, significantly higher than the 75% success achieved by classical methods [115]. The algorithm optimized pathfinding by dynamically adjusting fuzzy rules based on the motion error feedback, correcting trajectories more efficiently.

4.4. Technical Comparisons of Fuzzy Logic Approaches

A comparative assessment of fuzzy logic approaches reveals performance distinctions pertinent to adaptability, prediction capability, and motion correction are presented in Table 5:

Table 5. Comparison of fuzzy logic approaches.

Methodology	Key Features	Performance Benefits	Performance Limitations
Mamdani Fuzzy Model	Utilizes linguistic variables and fuzzy sets for inference.	Offers high interpretability and understanding.	Computationally intensive; less suitable for real-time applications.
Sugeno Fuzzy Model	Employs linear functions for output, faster computationally.	Improved efficiency and real-time application flexibility.	Reduced interpretability; less intuitive for human understanding.
Interval Type-2 Fuzzy	Captures a broader range of uncertainties.	Enhanced robustness to noise; superior accuracy under uncertainty.	High computational complexity and design difficulty.
Adaptive Neuro-Fuzzy	Combines neural networks and fuzzy logic for dynamic adjustments.	Capable of learning and adapting based on data inputs; effective in complex systems.	Requires large datasets and training time; prone to overfitting.
Hybrid Fuzzy Controllers	Combine fuzzy logic with PID, sliding Mode, etc.	Achieves improved stability and performance in control systems; reductions in error margins ranging from 25% to 40%.	Complex design and tuning; may increase implementation cost and computational load.

The advancements and adaptations in fuzzy logic architectures for cobot control have significantly impacted the field of robotics. By combining hybrid fuzzy-PID controllers, integrated predictive algorithms, and adaptive fuzzy techniques, researchers have achieved significant improvements over traditional control methods. Future work may focus on further integrating emerging AI approaches with fuzzy logic to enhance adaptability, precision, and overall efficiency in a wide range of robotic applications.

5. Main Research Questions

The control of collaborative robots (cobots) presents unique challenges, particularly in dynamic and uncertain environments. Fuzzy logic, known for its ability to manage vagueness and approximate reasoning, has emerged as a viable alternative to traditional control techniques.

The systematic literature review [116] was chosen as one of the most effective methods for stating our research topic. This section provides description of using fuzzy logic in cobots and considers the following research questions:

- How has fuzzy logic been applied to enhance cobot decision-making and adaptability in dynamic environments?
- How does fuzzy logic compare to other control methods (e.g., PID, neural networks, reinforcement learning) in terms of efficiency, accuracy, and computational complexity for cobot applications?
- What role does fuzzy logic play in human–robot collaboration, particularly in ensuring safety, flexibility, and intuitive interaction?

5.1. Application of Fuzzy Logic in Cobot Decision-Making and Adaptability

Fuzzy logic has been effectively employed to enhance cobot adaptability and decision-making in rapidly changing environments. Its strength lies in its capacity to model uncertainty and imprecision without requiring a precise mathematical representation of the system. In complex settings where cobots must interpret vague sensor inputs or respond to irregular human movements, fuzzy controllers provide smooth and robust responses [117]. For instance, adaptive fuzzy controllers have enabled mobile cobots to make real-time navigational decisions, balancing task execution with human presence in shared spaces [118].

Fuzzy logic is effectively used to enhance the adaptability and decision-making capabilities of collaborative robots (cobots) operating in rapidly changing and uncertain environments. Its main advantage lies in its ability to model uncertainty and imprecision without requiring an exact mathematical representation of the controlled system. This makes fuzzy logic particularly useful in scenarios where cobots must interpret vague sensor data or respond to unpredictable human actions [119].

In complex settings, such as manufacturing facilities or medical institutions, fuzzy controllers enable cobots to deliver smooth and stable responses. For example, adaptive fuzzy regulators have been applied to mobile cobots to support real-time navigation decisions while accounting for the presence of humans in shared workspaces [120].

Additional examples of fuzzy logic application in collaborative robotics illustrate its versatility across various operational contexts. In assembly and handling tasks, cobots equipped with fuzzy controllers can adaptively adjust grip strength or contact force based on object sensitivity, spatial positioning, and observed operator behavior [121].

Furthermore, fuzzy logic enables real-time adjustments of a cobot's velocity and path by evaluating proximity to obstacles, object density in the surrounding environment, and levels of sensor noise [122]. Even in suboptimal conditions, such as poor lighting or partial visual occlusion, fuzzy systems can interpret data from vision systems and depth sensors to ensure reliable manipulation and navigation [123]. For soft or multi-segmented manipulators, fuzzy logic facilitates the achievement of target shapes or positions by continuously compensating for structural deviations, thereby eliminating the need for complex inverse kinematic computations [124].

Thus, fuzzy logic demonstrates considerable potential in improving the reliability, safety, and efficiency of cobots operating in complex and dynamically changing environments. It is particularly valuable in situations where strictly formalised mathematical models of human behavior or the external environment are unavailable or impractical.

5.2. Comparison with Classical and AI-Based Control Methods

When compared with classical methods like PID controllers, fuzzy logic offers superior flexibility in nonlinear and uncertain conditions. While PID systems are computationally efficient and straightforward to implement, they struggle in situations involving abrupt changes or poorly modelled dynamics. Fuzzy logic, by contrast, accommodates uncertainty and allows for intuitive rule-based control without requiring detailed mathematical models. Neural networks and reinforcement learning (RL) approaches can learn complex behaviors but are typically computationally intensive and lack interpretability [125].

Several comparative studies highlight the advantages of fuzzy logic over traditional control methods in various applications. For instance, ref. [126] reported that a fuzzy controller exhibited superior performance compared to a PID controller in terms of convergence speed and robustness when operating under spatial constraints, making it a more suitable option in such environments. Similarly, ref. [127] demonstrated that, when applied to boost converters, fuzzy logic controllers offered better disturbance rejection and adaptability to changes in irradiance levels than their PID counterparts.

In experiments conducted by [128], fuzzy control was found to significantly outperform PID in maintaining response smoothness and stability, particularly under communication delay conditions. Furthermore, ref. [129] showed that fuzzy logic provided smoother and more accurate joint motion control than PID, especially in scenarios characterised by load variations and signal noise.

These findings collectively suggest that fuzzy logic offers a favorable balance between interpretability, robustness, and computational efficiency. As such, it remains a compelling

choice for control scenarios where classical or AI-based methods either fall short in flexibility or introduce unnecessary complexity.

5.3. Role in Human–Robot Collaboration

Fuzzy logic plays a vital role in ensuring both safety and intuitive interaction in human–robot collaboration (HRC) [130,131]. By using linguistic variables to model human-like reasoning, fuzzy controllers are able to predict and respond to user intentions, even in the presence of incomplete or ambiguous data. Studies have demonstrated that fuzzy control significantly enhances user trust, the fluidity of interaction, and the interpretability of the system, particularly in tasks involving physical contact and shared environments.

Recent advancements demonstrate how fuzzy logic can enhance human–robot collaboration by improving interpretability and responsiveness in real-world tasks. A notable development by [132] integrated fuzzy logic with video and speech language models to infer user intentions in real-time. This approach enabled collaborative robots to interpret natural commands more effectively and respond appropriately within dynamic and context-sensitive environments.

In another study, ref. [133] implemented fuzzy Q-learning techniques in cobots to regulate grip force and motion trajectories during cooperative manipulation tasks. These hybrid systems allowed robots to adapt to changes in user behavior more effectively, thereby reducing the risk of discomfort or injury during physical interaction.

Additionally, fuzzy logic has been applied to interpret vague temporal expressions—such as “soon,” “later,” or “in a moment”—in verbal instructions, allowing robots to better synchronize task execution with human expectations [134].

These applications illustrate fuzzy logic’s unique potential to bridge the gap between symbolic reasoning and physical execution in human–robot collaboration. By combining transparency with adaptability, fuzzy control systems support the development of safer, more intuitive, and human-centered robotic systems.

The reviewed literature clearly demonstrates the growing relevance and applicability of fuzzy logic in collaborative robot control. Across diverse research directions—ranging from adaptive decision-making and comparison with traditional control techniques to advancements in human–robot collaboration—fuzzy logic has shown significant advantages in handling uncertainty, improving system interpretability, and enabling responsive, human-centered interaction. Unlike classical controllers, fuzzy systems offer a robust framework for managing nonlinearity and imprecision, while remaining computationally efficient and explainable. Furthermore, their integration with machine learning and natural language processing opens new possibilities for intelligent and intuitive robotic behavior. Taken together, these findings establish fuzzy logic as a promising and versatile tool for enhancing the safety, adaptability, and performance of next-generation cobot systems.

6. Conclusions

The presented article reviews modern approaches to the application of fuzzy logic in collaborative robot control. The analysis demonstrates that fuzzy logic enhances robustness to uncertainty, increases the adaptability and flexibility of control systems, and improves the safety of human–robot interaction while ensuring intuitive operation.

From a scientific perspective, the development of fuzzy logic lies in its integration with artificial intelligence methods such as neural networks, evolutionary algorithms, and reinforcement learning. These hybrid approaches form the foundation for predictive systems capable of self-learning and generating their own rules to adapt to uncertain conditions and ensure coordinated actions between humans and robots.

In practice, fuzzy logic has already been applied in industry, including assembly, logistics, flexible manufacturing, and other processes. Its use reduces operational risks and equipment downtime while improving ease of use. The most significant potential lies in its application within mass customization environments, where the rapid and safe reconfiguration of robotic systems for changing tasks is required.

The literature review highlights several promising future directions for fuzzy logic development: automatic rule-based generation to reduce system tuning and calibration time; the use of neural networks to decrease algorithmic computational complexity; integration with digital twins and virtual testing systems; and the advancement of multi-agent fuzzy control systems. All these innovations aim to overcome the limitations of classical fuzzy logic-based control. Progress in this field, particularly through integration with other intelligent methods, can improve prediction accuracy, enhance system diagnostics, reduce computational load in real-time operations, and enable the coordination of multi-level systems with human operators—thus improving the transparency and reliability of fuzzy logic-based control and diagnostic frameworks.

In conclusion, fuzzy logic is a key element in shaping the next generation of collaborative robot control and diagnostic systems under uncertain conditions. Its continued development will enable the creation of human-centered control solutions designed to ensure safety and efficiency in future manufacturing environments.

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